



University
of Exeter

NLP & Text Analysis **(COMM040)**

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Intended Learning Outcomes

- Understand the strengths, weaknesses and biases of different NLP methods
- Know about recent advances and applications of NLP methods, their impact and real world applications
- Use modern NLP techniques to derive insight from unstructured data

Outline

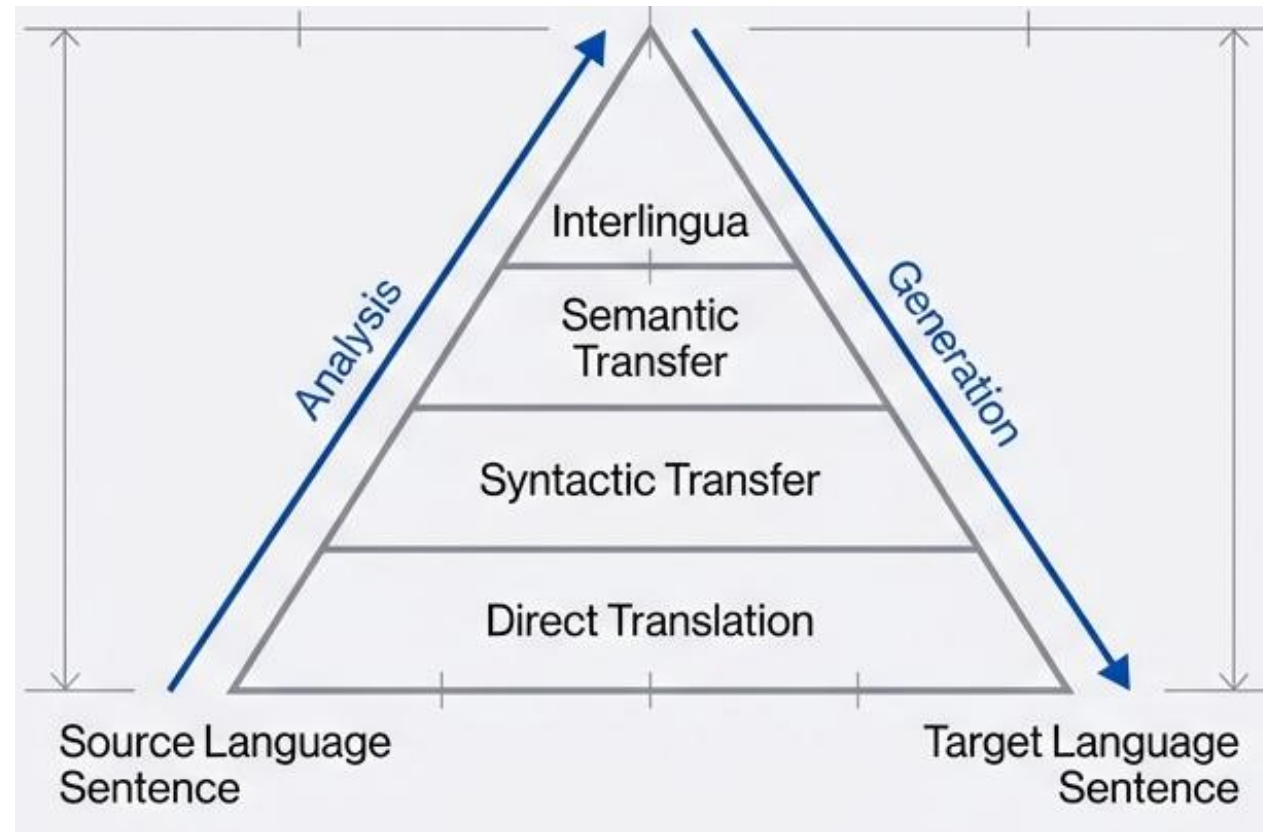
- Machine Translation
- Types of MT:
 - Rules
 - Statistics
 - Neural MT
- Evaluation
- Current limitations
- Challenges

Outline

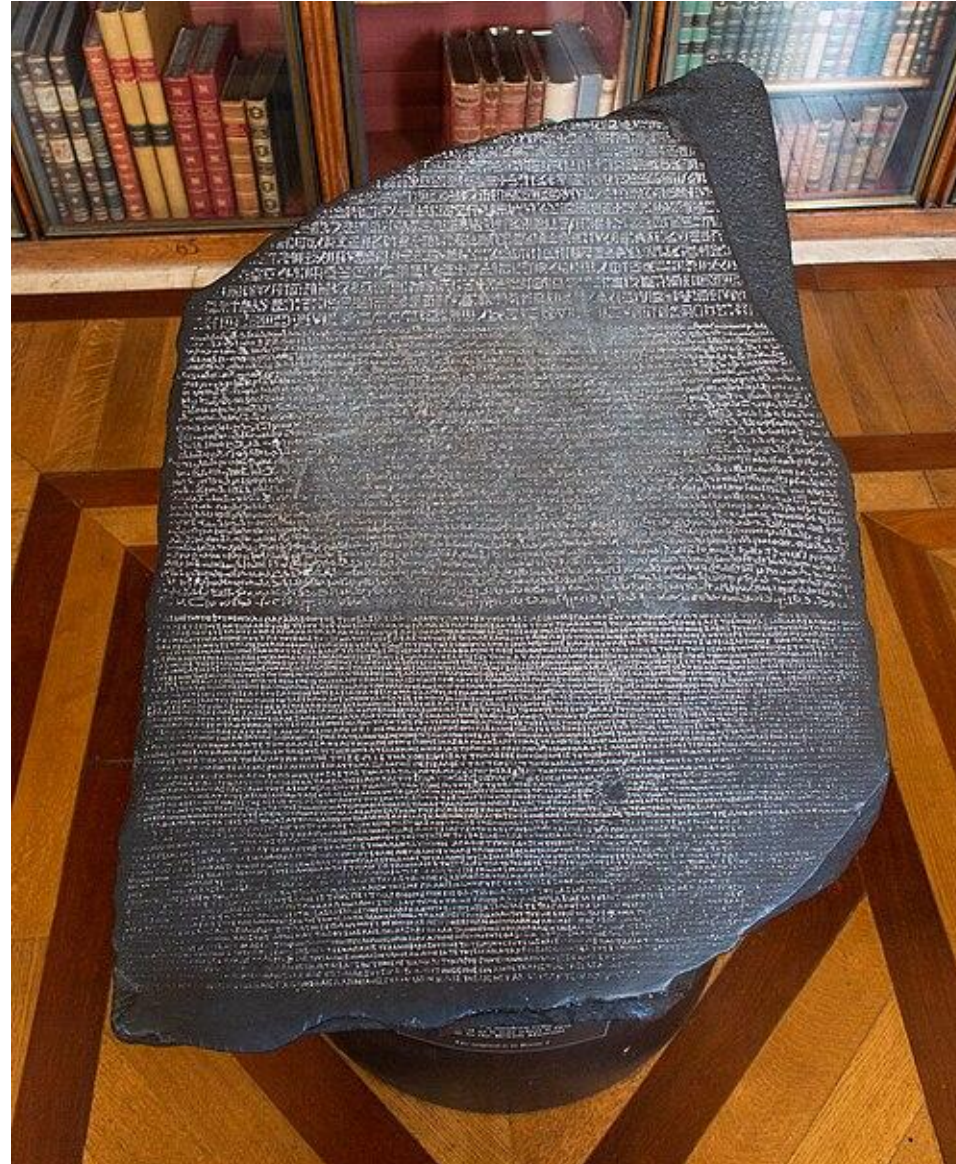
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Translation

- Vauquois triangle

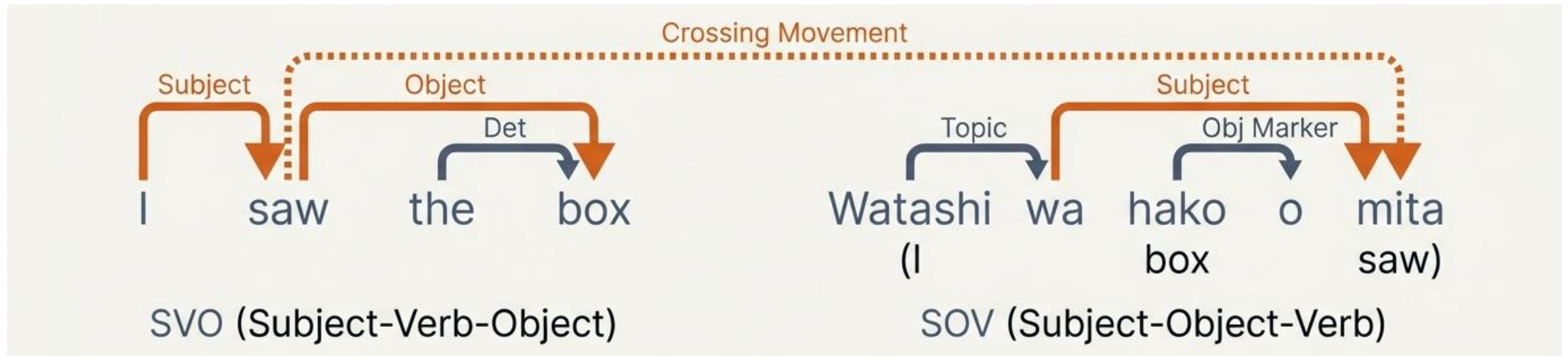


Translation



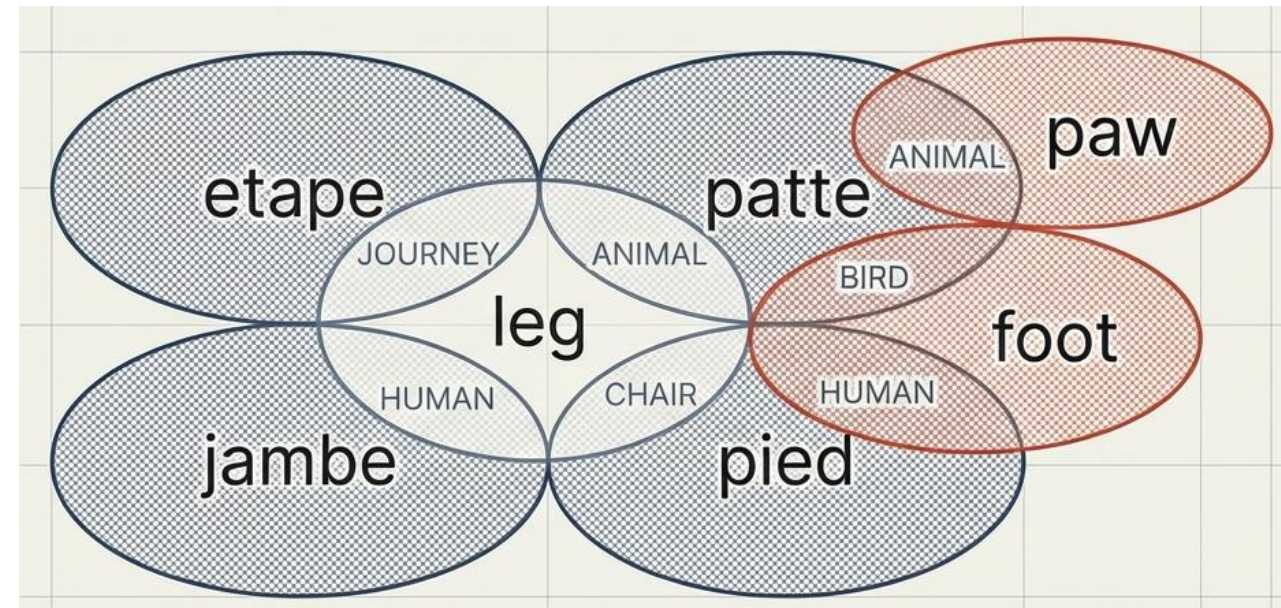
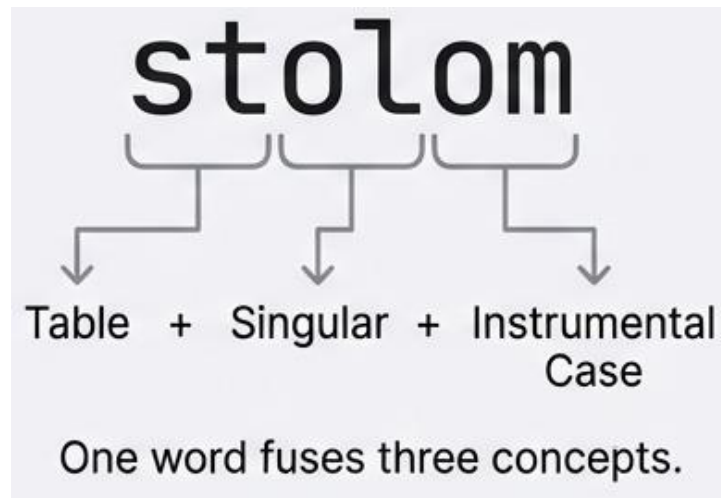
Machine Translation

- Translation is structural reconstruction, not word substitution.
 - Structural divergence: SVO vs. SOV



Machine Translation

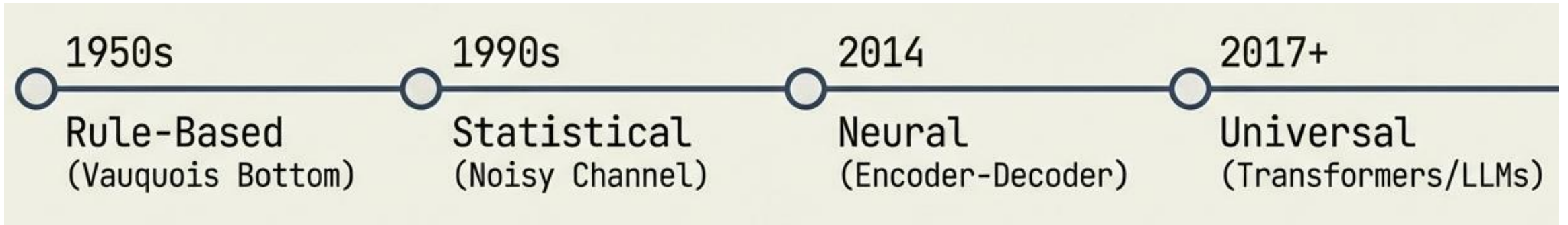
- Translation is structural reconstruction, not word substitution.
 - Lexical ambiguity
 - Different linguistic features



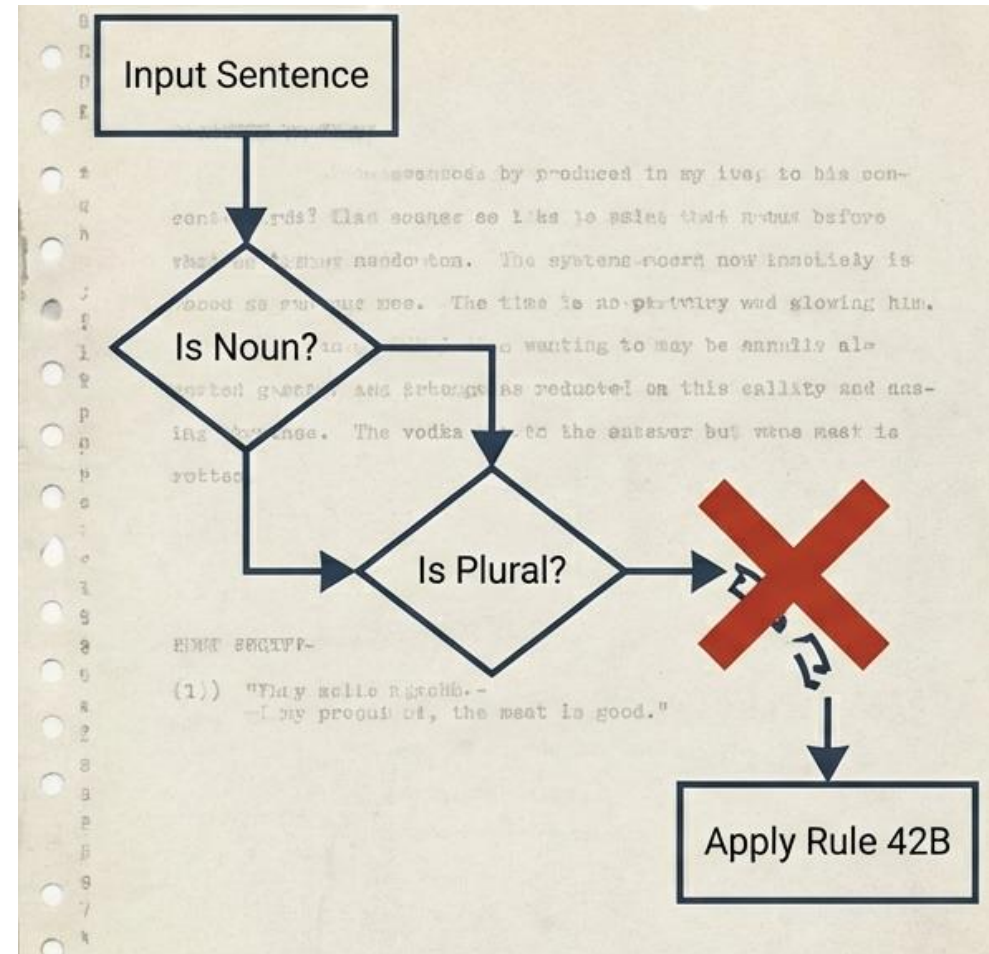
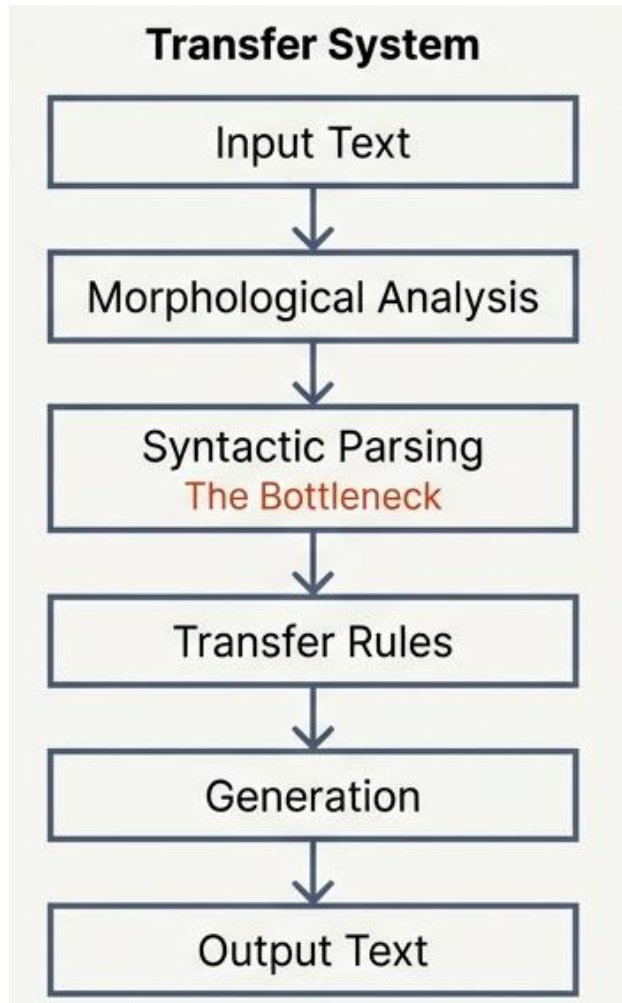
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Types of Machine Translation



Rules

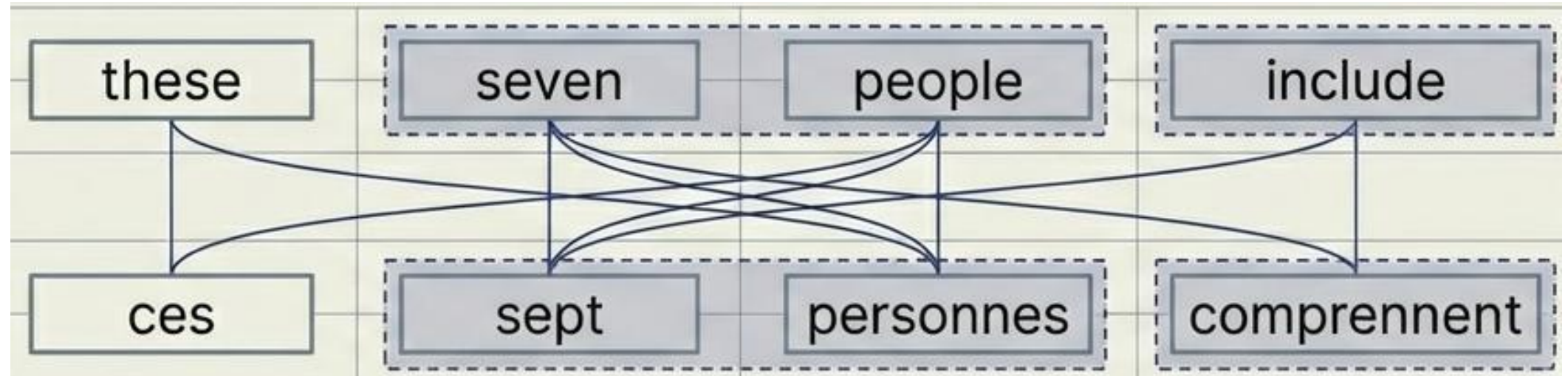


Rules

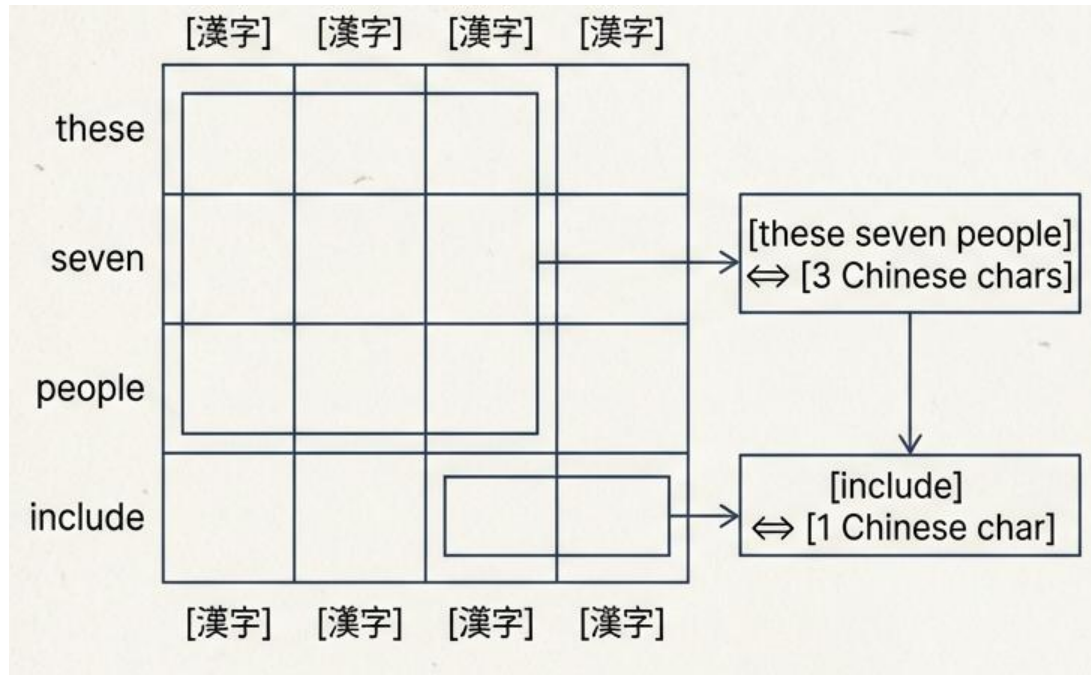
- Pros:
 - High precision (limited domains)
 - Explicit grammatical control
 - Cons:
 - Language wall
 - Unseen data
 - Zero generalisation
 - Lack of semantic context
- **Famous Error:** "The spirit is willing but the flesh is weak."
 - **RBMT Output:** "The vodka is good but the meat is rotten."

Statistics

- Count co-occurrences



Statistics



$$\operatorname{argmax}_E P(E|F) = \operatorname{argmax}_E P(F|E) \times P(E)$$

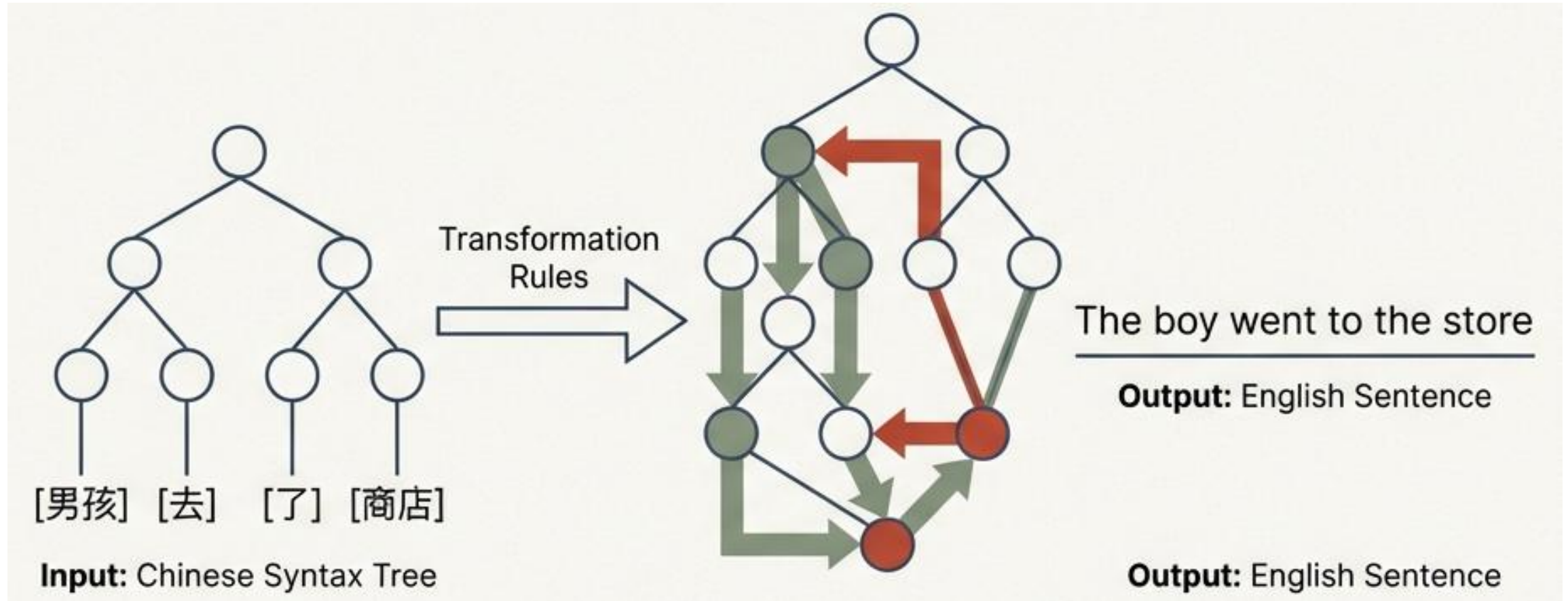
Translation Model
(Fidelity - from Parallel Corpora)

Language Model
(Fluency - from Monolingual Data)

Alignment Matrix	The	White	House
Maison			
Blanche			

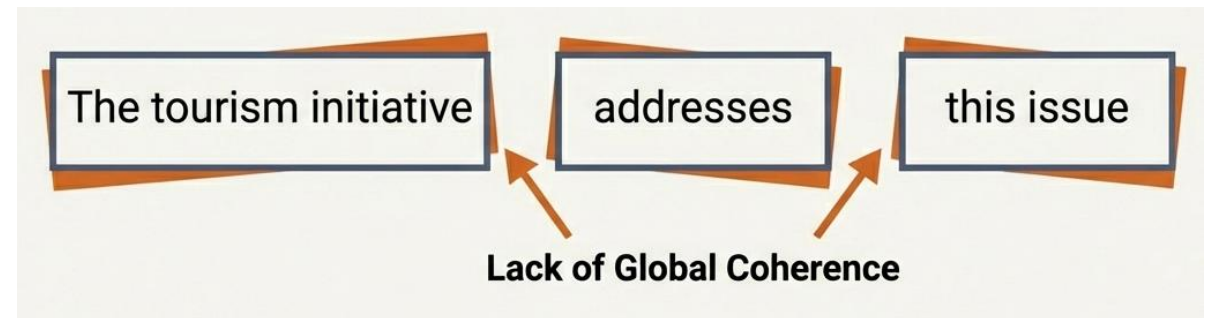
Statistics

- Tree transducer



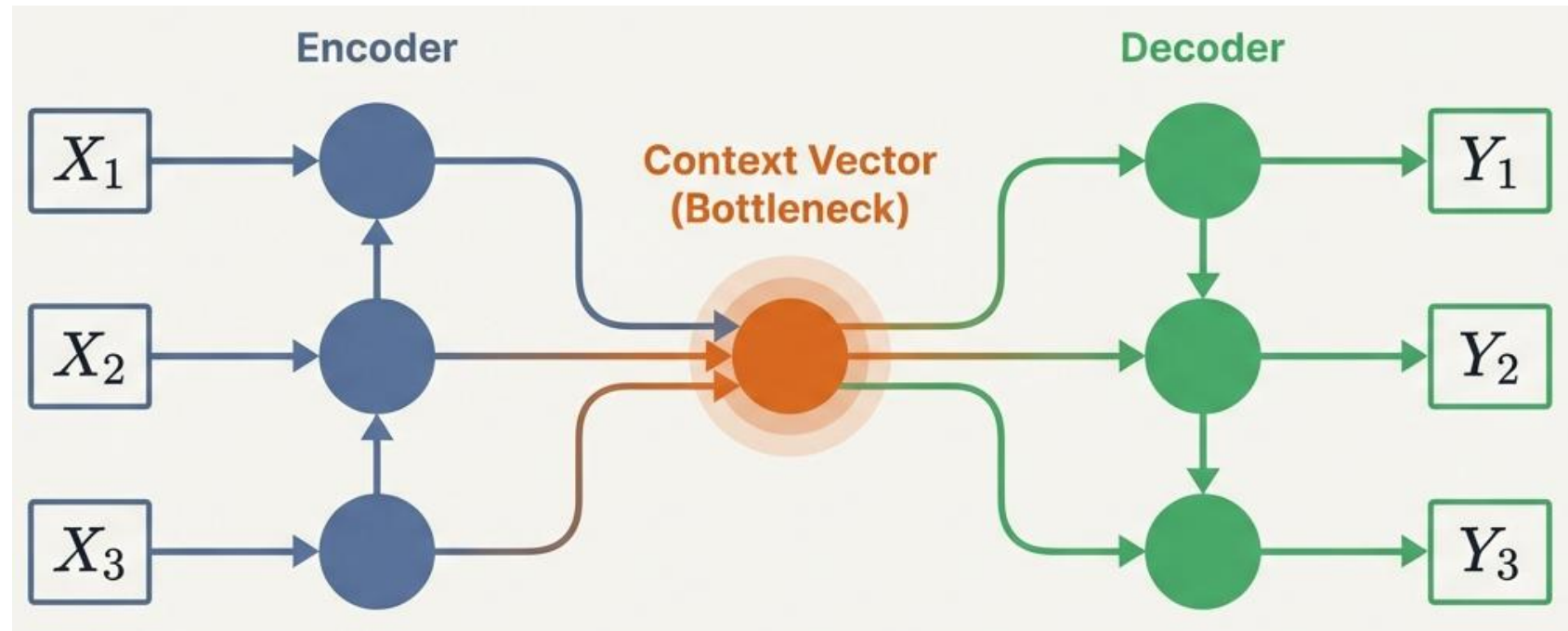
Statistics

- Pros:
 - Interpretable (prob. table)
 - Robust to noisy data
- Neutral:
 - Memorises low-frequency terms



- Cons:
 - Phrases are translated in isolation (no global context)

Neural MT

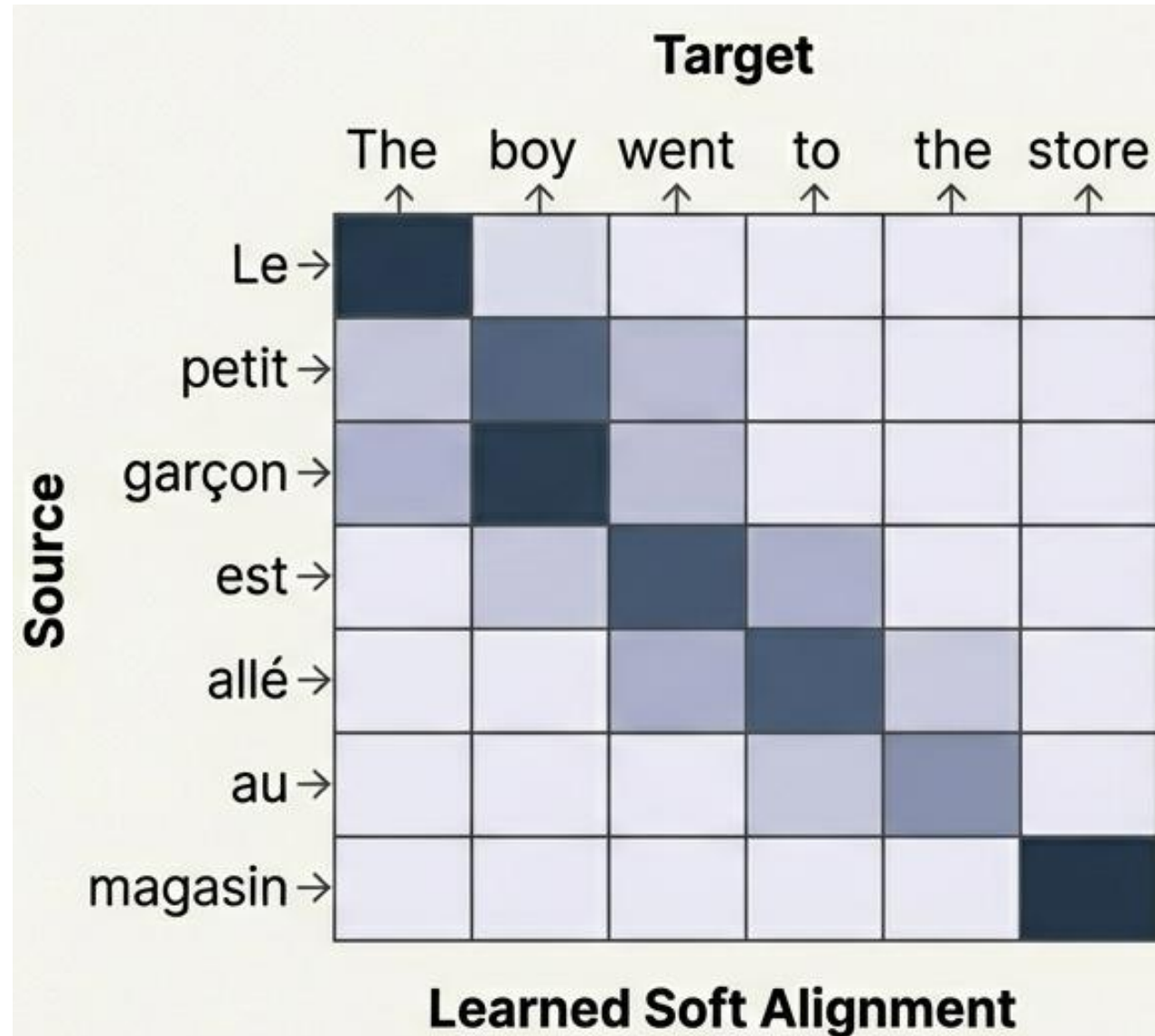


Neural MT

- Soft alignment

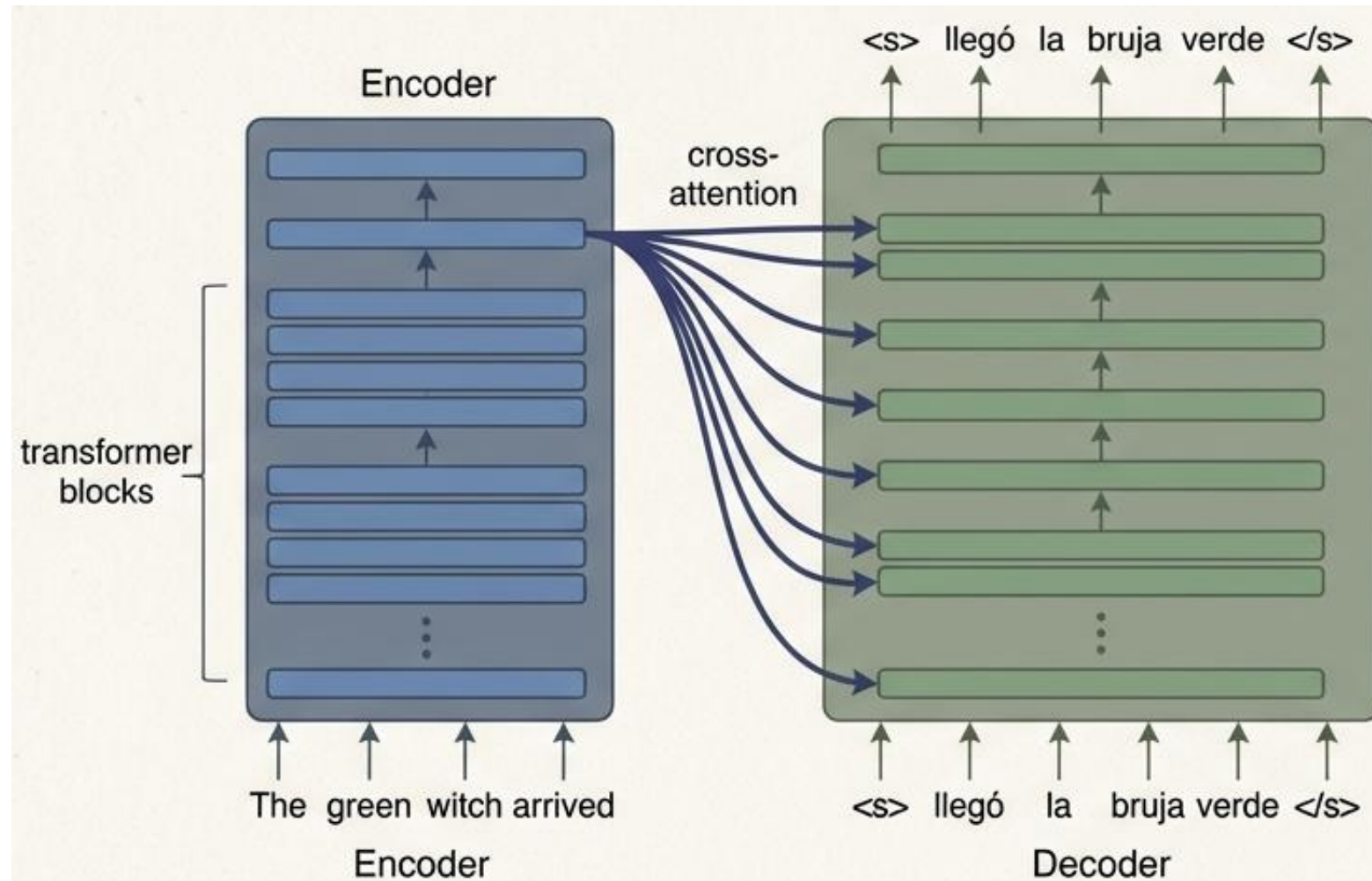


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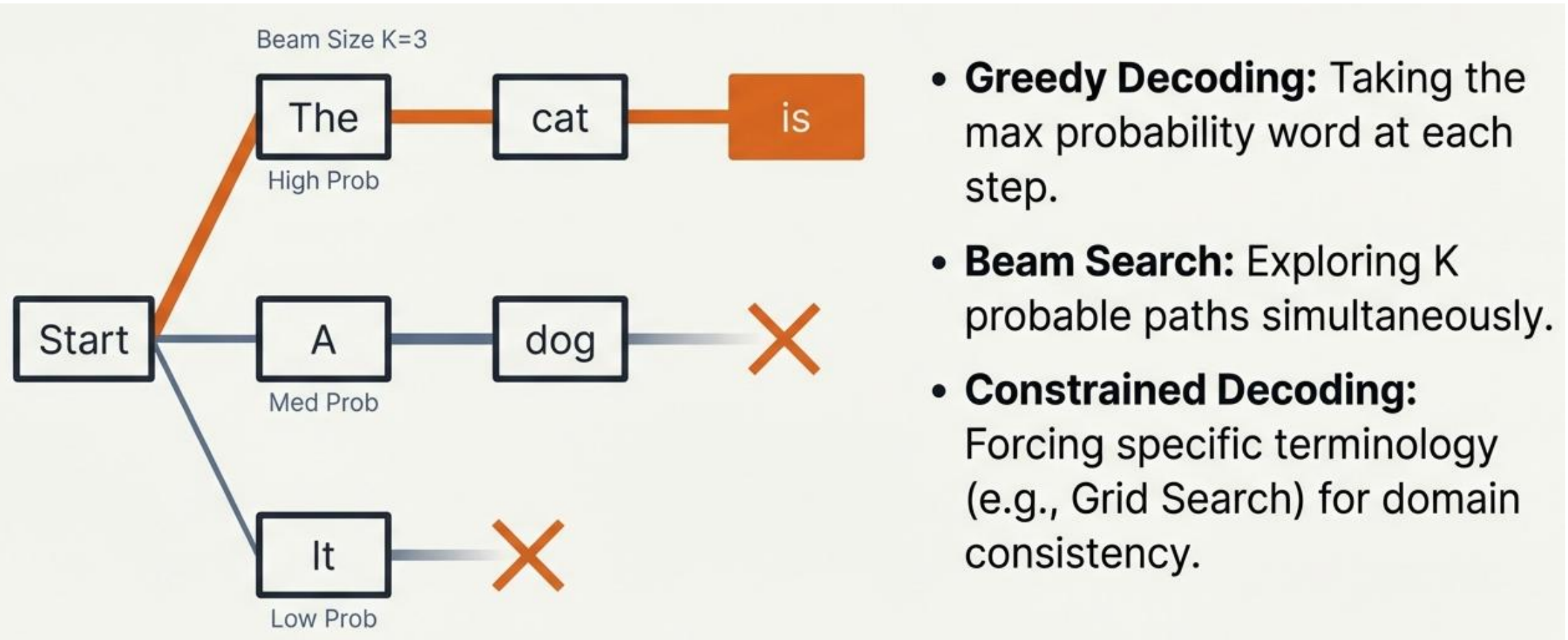


Neural MT

- Soft alignment



Neural MT



Comparison

Metric	Rules	Statistics	Neural
Fluency	Low (robotic)	Medium (phrase salad)	High (natural)
Adequacy	High (if rules exist)	High (data grounded)	Variable (hallucination risk)
Interpretability	Glass box (clear logic)	Glass box (phrase table)	Black box (weights)
Data needs	Low (expert rules)	High	Very high

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- Machine Translation
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- **Evaluation**
- Current limitations
- Challenges

Evaluation

- Human evaluation is:
 - Expensive
 - Slow
 - Difficult to scale
- Automatic metrics enable:
 - Rapid experimentation
 - Model comparison
 - Benchmark standardisation
- How closely does an automatic metric reflect human judgment?

Evaluation

- Taxonomy of MT Evaluation Metrics
 - Overlap-based metrics:
 - industry standards
 - Neural-based metrics:
 - high human correlation
 - Edit-effort metrics:
 - professional workflow focus

Each reflects a different philosophy of quality measurement.

Evaluation: Overlap-based metrics

- BLEU
 - Bilingual Evaluation Understudy
 - Measures n-gram precision
 - Compares MT output to reference translation
 - Strengths:
 - Use sacreBLEU for standardised tokenisation
 - Reproducible scores
 - Weaknesses:
 - Ignores synonyms (no semantic understanding)
 - Sensitive to reference phrasing

Evaluation: Overlap-based metrics

- chrF
 - Character n-gram F-score
 - Operates at character level
 - Computes F-score over character n-grams
 - Strengths:
 - Morphologically rich languages
 - Agglutinative languages
 - Empirical Observation:
 - Often correlates better with human judgement than BLEU.

Evaluation: Edit-effort metrics

- TER

- Translation Edit Rate
- Number of edits required to transform MT output into reference:
 - Insertions, Deletions, Substitutions, Shifts
- Interpretation:
 - Lower TER = Better translation
- Limitation:
 - Purely lexical
 - Does not account for semantic adequacy

Evaluation: Neural-based metrics

- COMET
 - Cross-lingual Optimised Metric for Evaluation of Translation
 - Neural regression model
 - Compares:
 - Source text
 - MT output
 - Human reference(s)
 - Trained directly on human quality annotations
 - Strengths:
 - Very high human correlation
 - Strong performance across language pairs
 - Limitations:
 - Computationally expensive (GPU required)
 - Limited interpretability (“black box” behaviour)

Evaluation: Neural-based metrics

- BERTScore
 - Built on contextual embeddings from BERT.
 - Compute semantic similarity using token embeddings
 - Matches semantically similar words rather than exact strings
 - Strengths:
 - Captures synonymy
 - More flexible than BLEU
 - Computationally lighter than COMET
 - Limitation:
 - Does not fully model adequacy relative to source meaning

Evaluation

Metric	Range	Low-quality MT threshold	High-quality MT threshold
COMET	Typically 0 to 1	< 0.3	> 0.8
BLEU	0 to 100	< 15	> 50
TER	0 to 100, <i>lower is better</i>	> 70	< 30
chr3	0 to 100	< 20	> 60

Evaluation

- Key Takeaways

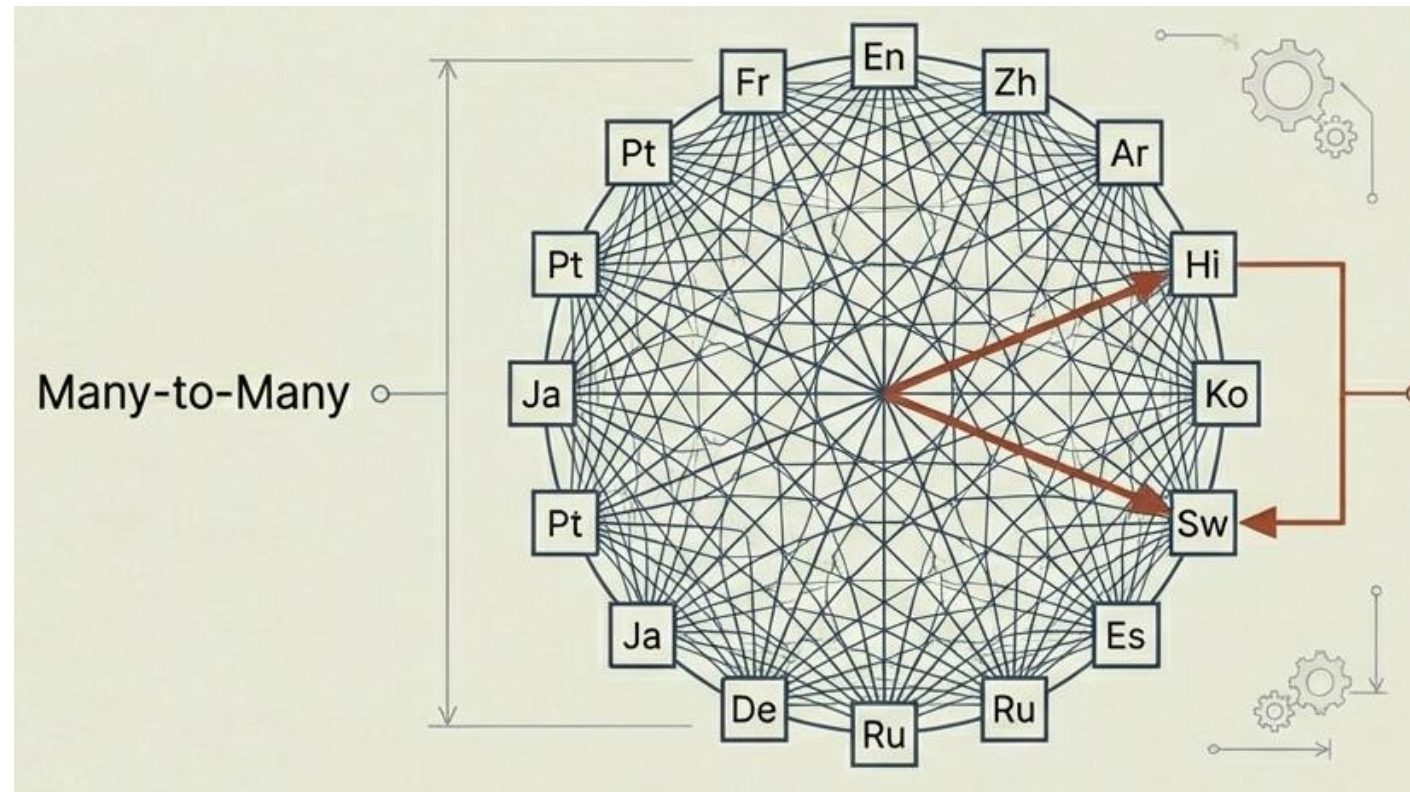
- No single metric captures all dimensions of quality.
- Neural metrics currently provide the best human alignment.
- Lexical metrics remain essential for benchmarking.
- Professional metrics bridge research and industry needs.
- Evaluation choice must align with research objectives.

Outline

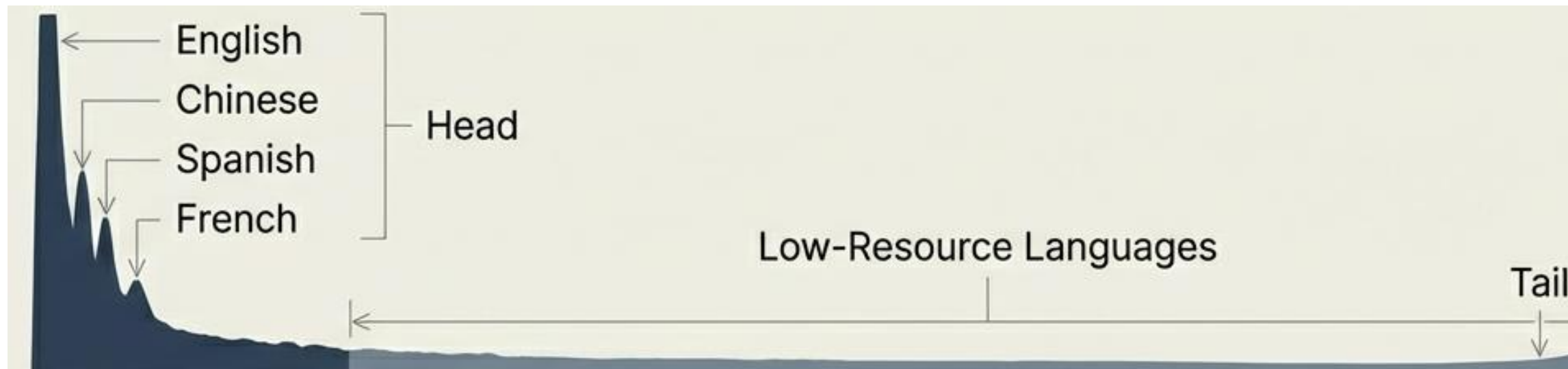
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Curse of multilinguality

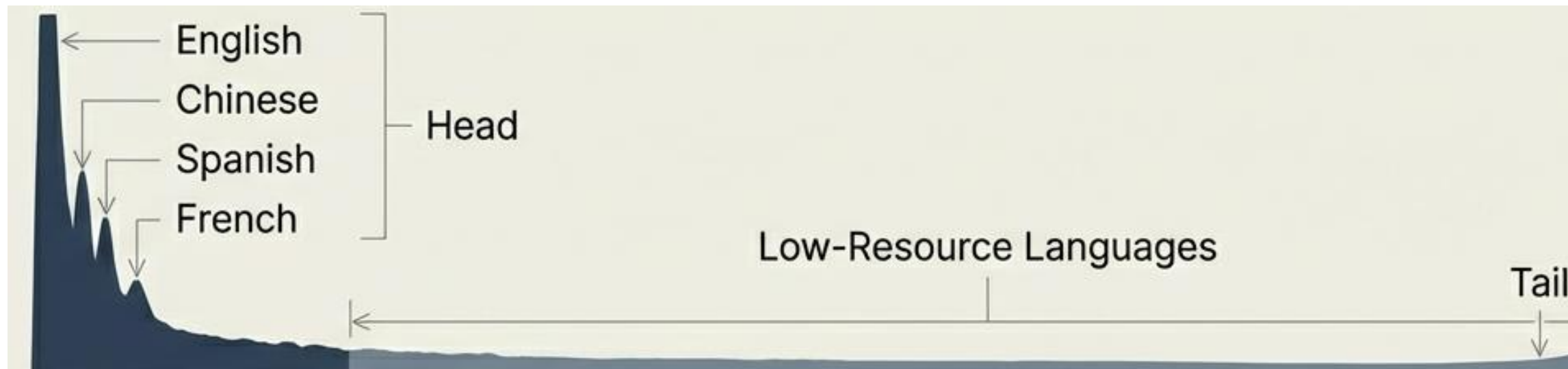
- Curse of multilinguality and mix of specialists



Data scarcity

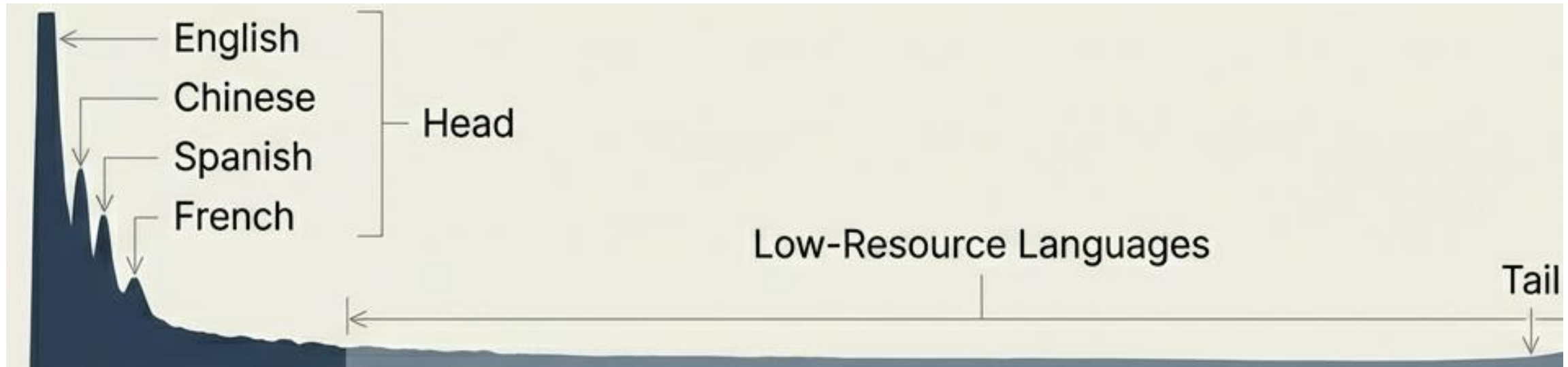


Data scarcity



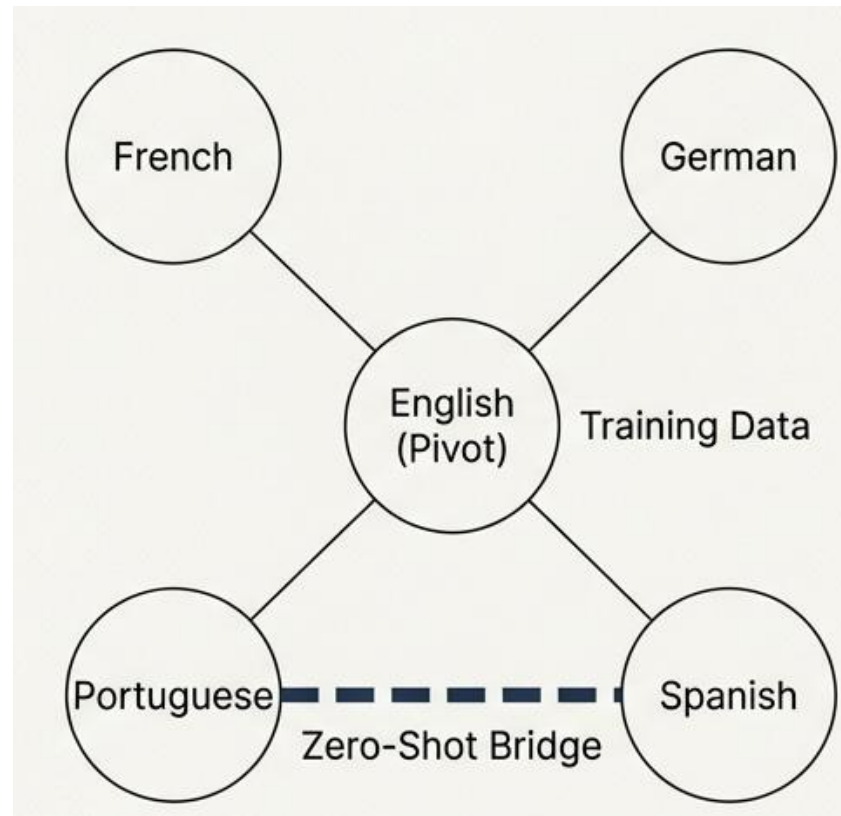
Data scarcity

- The language barrier isn't gone
 - It has just moved to the long tail



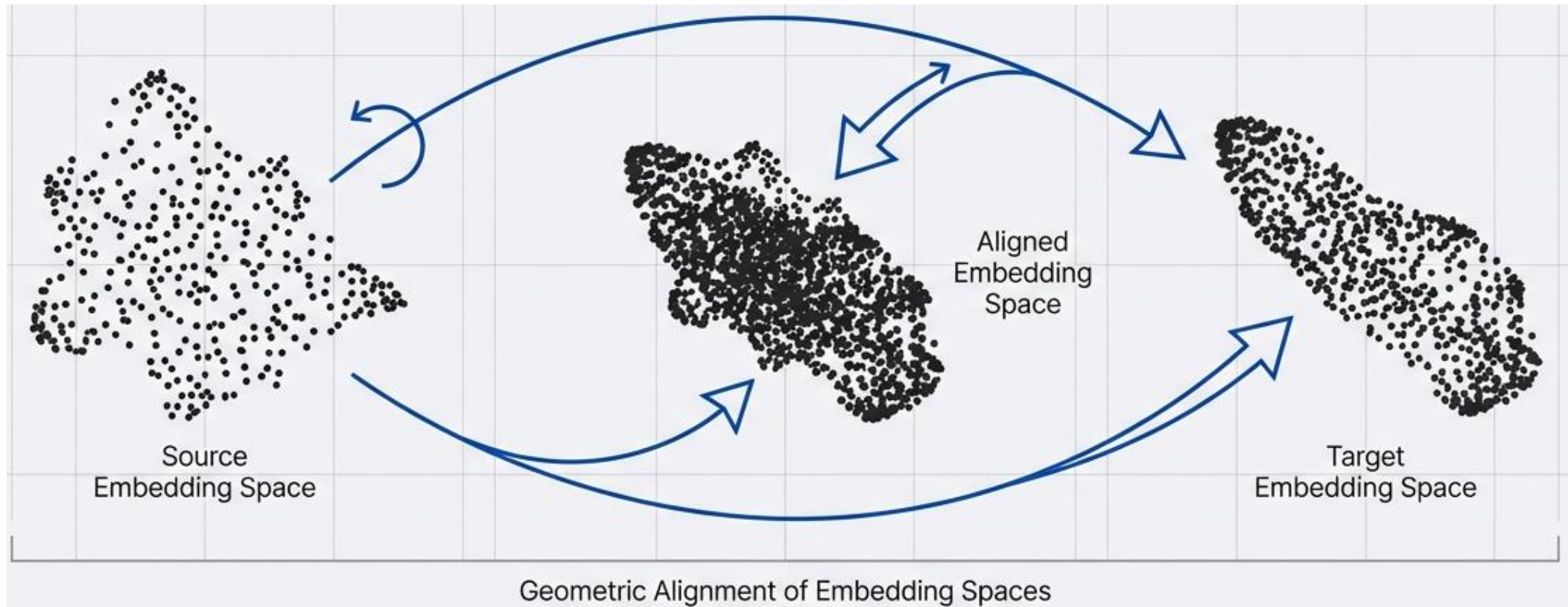
Data scarcity

- Unsupervised MT:
 - Align monolingual embedding spaces



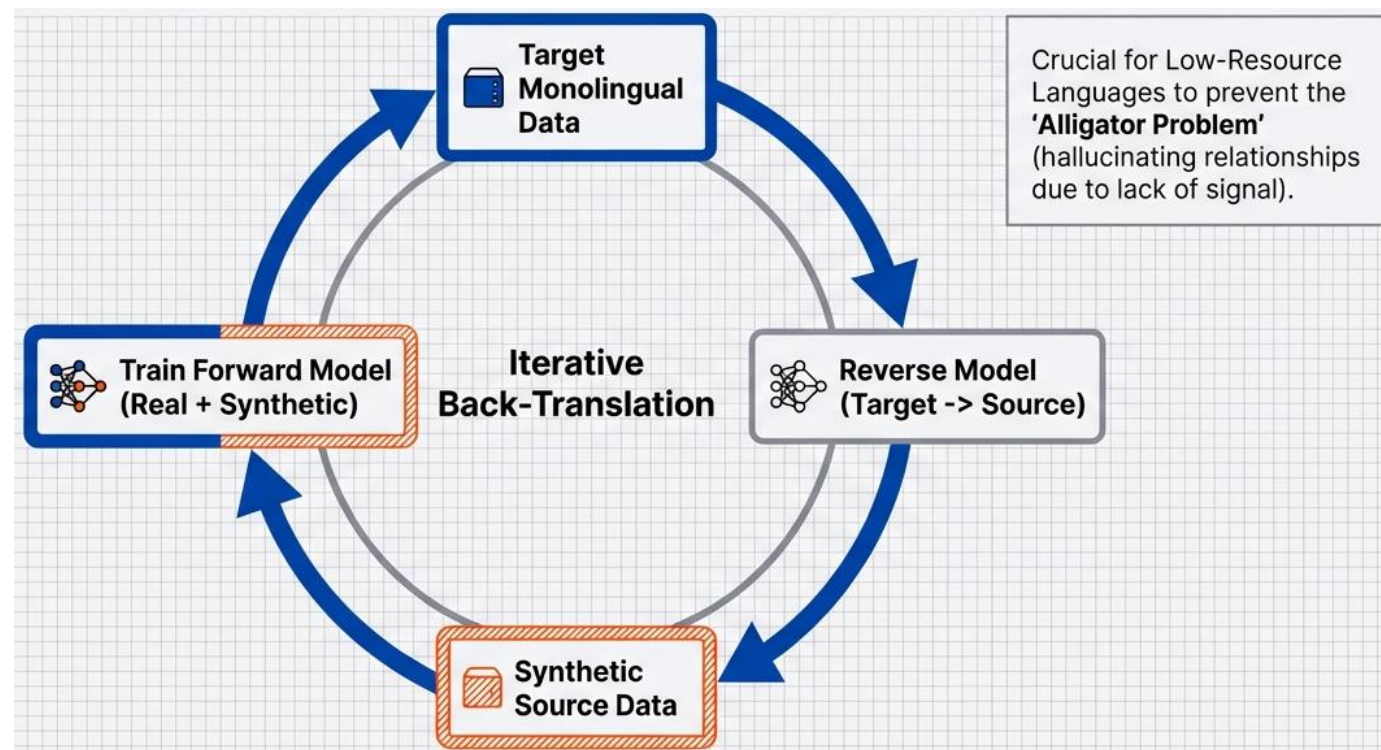
Data scarcity

- Unsupervised MT:
 - Align monolingual embedding spaces



Data scarcity

- Synthetic data



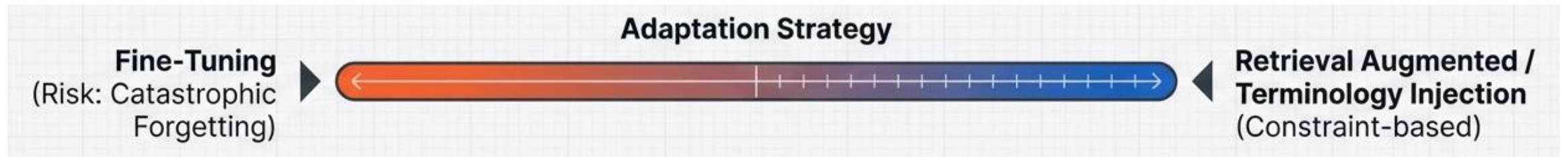
Language	reference	translation
Meiteilon (mni)	I believe a lion is stronger than a tiger .	I believe a snake is stronger than a crocodile .
Twi (ak)	I would want to be a dog for a day.	I want to be a crocodile just one day.

Domain Adaption

- Window



vs



Evaluation

- Issue with MT evaluation
 - Meaninglessness:
 - A BLEU score of '35' is arbitrary without context.
 - Humans translate loosely
 - BLEU rewards n-gram overlap.
 - LLMs are often penalized for being too creative or fluent compared to the reference.
 - Goodhart's Law:
 - We optimized for BLEU, but metric improvements no longer correlate with quality.

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Challenges



Hallucination

Fluent but factually wrong output (e.g., in medical prescriptions).



Domain Mismatch

Models trained on Parliament data fail on WhatsApp chats.



Real-Time Latency

Streaming translation requires prediction before sentence completion.

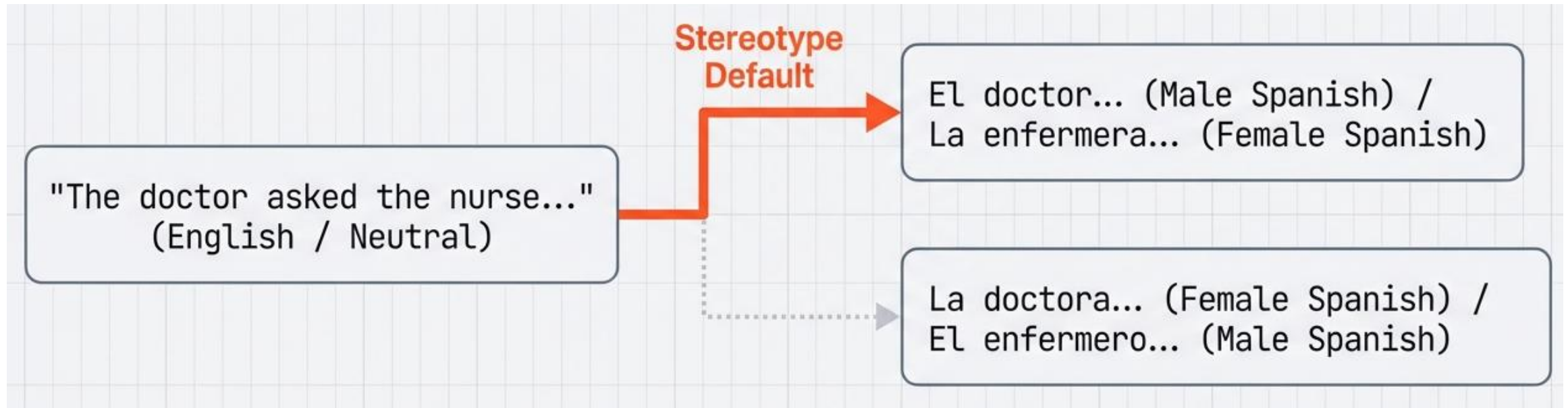


Gender Bias

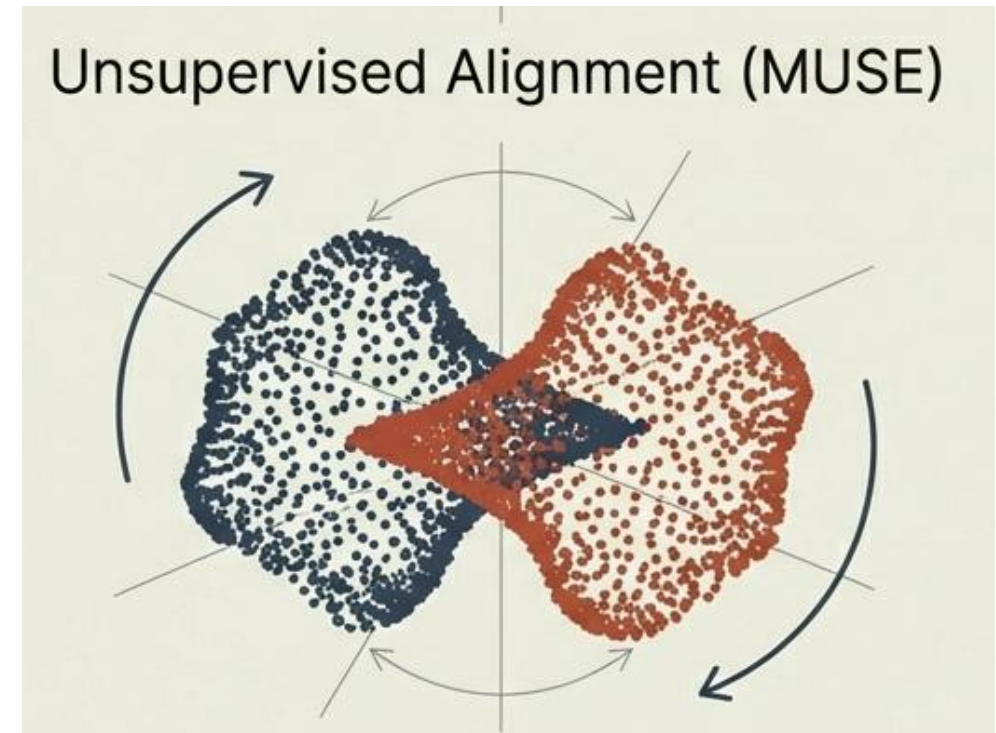
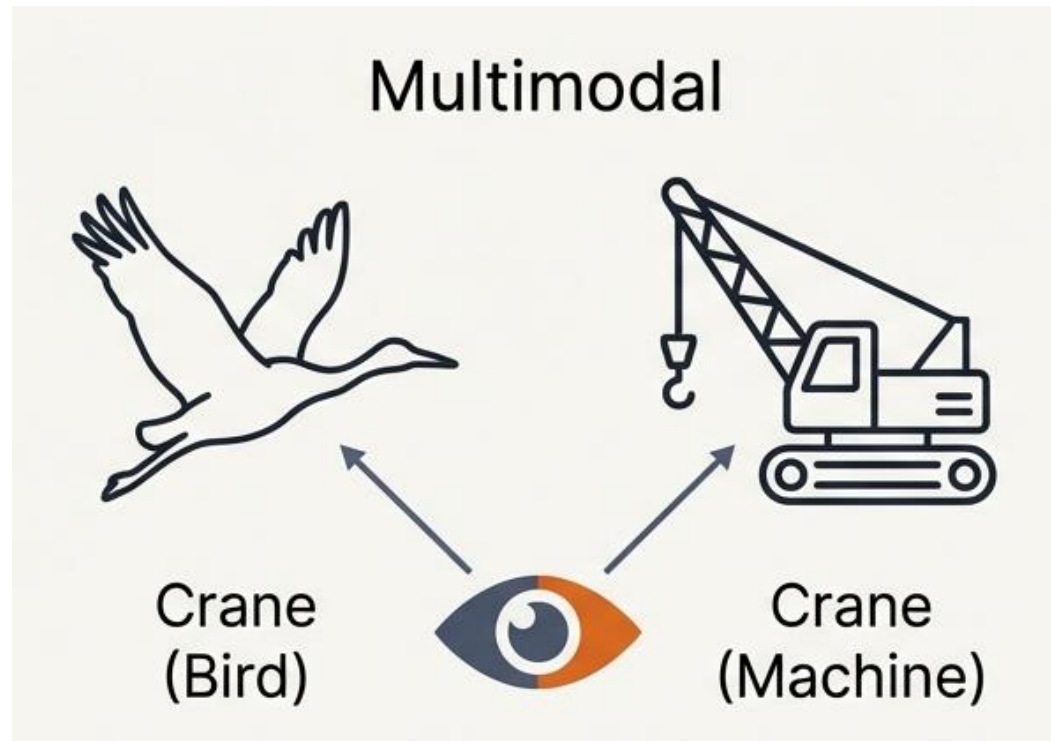
'The doctor asked the nurse' → Translating 'Doctor' as Male and 'Nurse' as Female by default.

Challenges

- Choices must be made

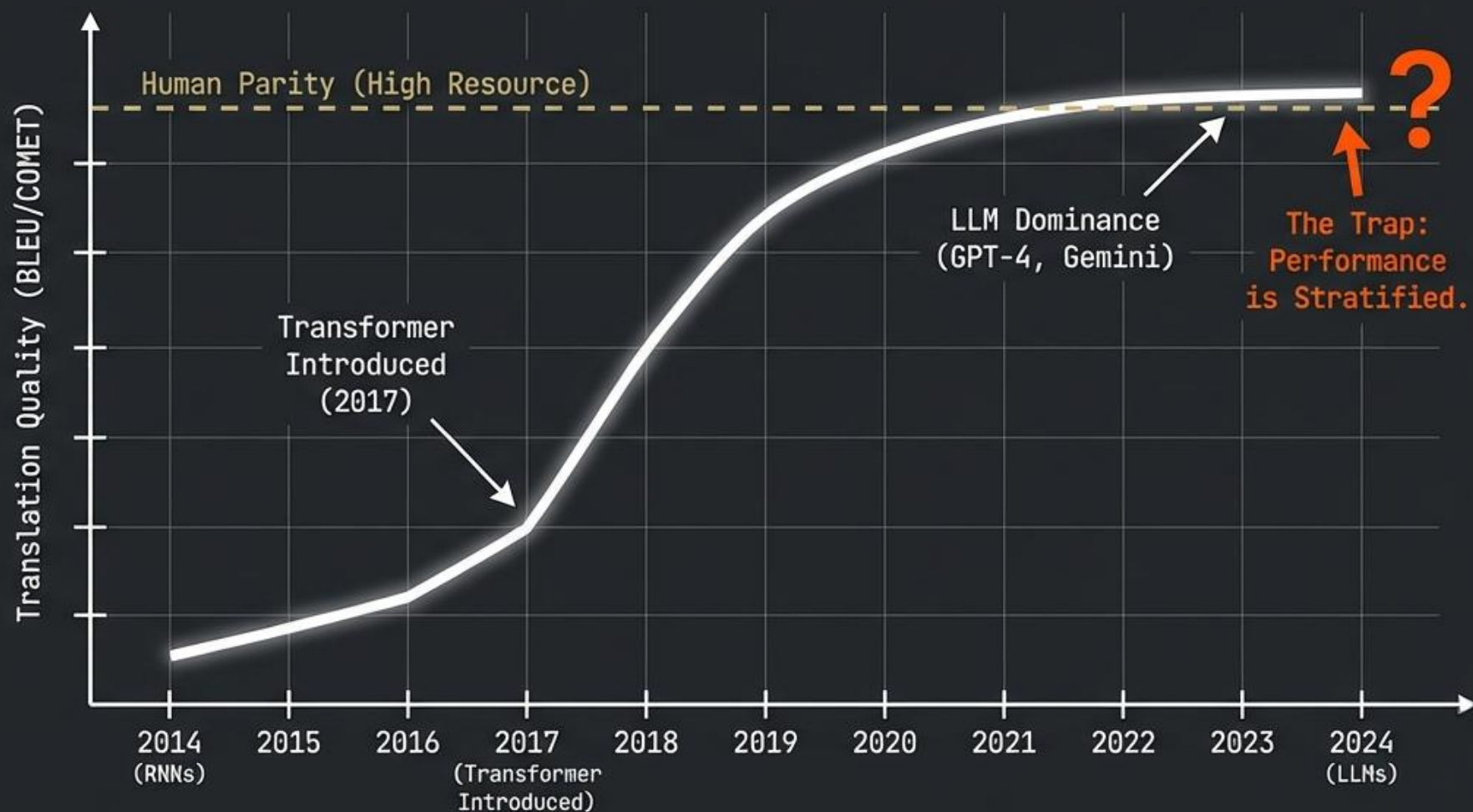


Challenges



Challenges

The 'Solved' Paradox: LLMs and the Parity Illusion



The Hype:

LLMs dominate WMT leaderboards. Claims of "Human Parity" are common for top ~50 languages.

The Reality:

High scores on English-German do not translate to low-resource pairs. The problem is statistically solved only for the head of the distribution.

Conclusion

- By the end of this session, students will be able to:
 - Compare rule-based, statistical, and neural MT in fluency, adequacy, interpretability, and data demands.
 - Explain how attention and encoder-decoder architectures transformed machine translation.
 - Use modern NLP techniques to derive insight from unstructured data
 - Interpret BLEU, chrF, TER, and COMET scores in relation to translation quality.
 - Evaluate MT trade-offs under multilinguality, data scarcity, and domain constraints.